

Summary Report



1. Strengths and weaknesses of each classifier

- **KNN**
 - **Strengths:** Simple to understand, works well when classes are well separated, and achieved perfect performance on this dataset.
 - **Weaknesses:** Sensitive to the choice of k and distance metric; can become slower at prediction time as the dataset grows.
- **SVM**
 - **Strengths:** Strong margin-based classifier, effective for high-dimensional data, and performed very well with both linear and non-linear kernels.
 - **Weaknesses:** Less interpretable than simple models, and performance depends on choosing an appropriate kernel and regularization parameter.
- **Naive Bayes**
 - **Strengths:** Fast, simple, probabilistic, and easy to interpret. It also performed very well on this dataset.
 - **Weaknesses:** Assumes feature independence, which may not hold in real-world message data, so it may be less accurate when features are strongly correlated.

2. Impact of probability assumptions in Naive Bayes

Naive Bayes relies on the assumption that input features are conditionally independent given the class. In this project, this assumption simplifies the probability calculation and makes the model efficient. However, if the engineered message features are correlated (for example, message length and word count), the model may overestimate or underestimate posterior probabilities. Even so, the model still performed well here, showing that the assumption was not too harmful for this dataset.

3. Trade-offs between interpretability and performance

- **KNN** is easy to explain but less efficient for large datasets.
- **SVM** often gives strong predictive performance but is less intuitive because its decision boundary is based on support vectors rather than directly interpretable rules.
- **Naive Bayes** is the most interpretable because its predictions can be explained through conditional probabilities, though its independence assumption may reduce flexibility.

4. Business recommendation for real-world deployment

For a real-world spam detection system, a **hybrid approach** is recommended. A model such as **SVM** or **KNN** can be used as the main classifier for strong predictive performance, while **Naive Bayes** can serve as a lightweight baseline or explainability tool. If the business values transparency and fast deployment, Naive Bayes is a strong starting point. If the priority is maximum accuracy and robust separation, SVM is the better production choice.

5. Final conclusion

All three classifiers achieved excellent results on this dataset. Based on the experiments, **SVM and KNN** are strong candidates for deployment, while **Naive Bayes** is valuable for its simplicity, speed, and probabilistic interpretability.